Lesson 20: Projections, Orthonormal Bases, and Riesz Representation

AI/ML Mathematics

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1 Motivation and Overview

In Lesson 19 we equipped vector spaces with inner products, establishing the geometry of angles and orthogonality. We now develop the three main consequences of this geometry in the Hilbert space setting: orthogonal projections, Fourier expansions in orthonormal bases, and the Riesz Representation Theorem. Together, these results provide the theoretical machinery for least-squares approximation, spectral decomposition, and the identification of a Hilbert space with its dual—all of which are fundamental to function approximation in reinforcement learning.

Projections and Representation in RL

1. **Projected Bellman equation:** In LSTD (Least-Squares Temporal Difference), the value function is approximated by solving $\hat{V} = \Pi_\mu T\hat{V}$, where Π_μ is the orthogonal projection onto a feature subspace in the $L^2(\mu)$ inner product. The projection theorem guarantees existence and uniqueness of $\Pi_\mu T\hat{V}$.
2. **Fourier features:** Random Fourier features approximate kernel functions via truncated Fourier expansions. Parseval's identity provides the theoretical justification: the truncation error equals the tail of the Fourier coefficient series.
3. **Riesz representation and gradients:** The Riesz theorem identifies bounded linear functionals with inner product representatives. This underpins the definition of the gradient: $Df(x)[h] = \langle \nabla f(x), h\rangle$, and more generally the natural gradient via the Fisher information inner product.

Prerequisites. Inner product spaces, Hilbert spaces, orthogonality, orthogonal complements (Lesson 19). Completeness and Banach spaces (Lessons 15–16).

Outline. Section 2 lists references. Section 3 proves the projection theorem and develops orthogonal projections. Section 4 introduces orthonormal systems, the Gram–Schmidt process, and Bessel's inequality. Section 5 establishes Parseval's identity and Fourier expansions. Section 6 connects separability to countable orthonormal bases. Section 7 proves the Riesz Representation Theorem. Section 8 provides exercises. Section 9 summarizes the key results.

2 References

The projection theorem and Riesz representation are covered in Kreyszig [1989], Chapters 3.3–3.6; Conway [1990], Chapter I.2–I.5; Brézis [2011], Chapter 5; and Rudin [1991], Chapter 4. The treatment of orthonormal bases and Parseval's identity follows Lax [2002], Chapter 6, and Kreyszig [1989], Chapter 3.5.

3 The Projection Theorem

Theorem 3.1 (Best Approximation in Closed Convex Sets)**.** *Let H be a Hilbert space and $C \subset H$ a nonempty, closed, convex set. For every $x \in H$, there exists a unique $y_0 \in C$ such that*

$$\|x - y_0\| = \inf_{y\in C} \|x - y\| =: \operatorname{dist}(x, C).$$

Proof. Let $d = \inf_{y\in C} \|x - y\|$. Choose a minimizing sequence (y_n) in C with $\|x - y_n\| \to d$.

Claim: (y_n) is Cauchy. By the parallelogram law applied to $a = x - y_n$ and $b = x - y_m$:

$$\|a + b\|^2 + \|a - b\|^2 = 2(\|a\|^2 + \|b\|^2).$$

Here $a - b = y_m - y_n$ and $a + b = 2x - y_n - y_m = 2\big(x - \frac{y_n+y_m}{2}\big)$. Since C is convex, $\frac{y_n+y_m}{2} \in C$, so $\|a+b\| = 2\big\|x - \frac{y_n+y_m}{2}\big\| \geq 2d$. Therefore

$$\|y_n - y_m\|^2 = 2(\|x-y_n\|^2 + \|x-y_m\|^2) - 4\left\|x - \frac{y_n+y_m}{2}\right\|^2 \leq 2(\|x-y_n\|^2 + \|x-y_m\|^2) - 4d^2.$$

As $n, m \to \infty$, $\|x-y_n\|^2 \to d^2$ and $\|x-y_m\|^2 \to d^2$, so $\|y_n - y_m\|^2 \to 0$. Thus (y_n) is Cauchy.

Since H is complete and C is closed, $y_n \to y_0 \in C$, and $\|x - y_0\| = \lim \|x - y_n\| = d$.

Uniqueness: If y_0, y_0' both achieve the infimum, apply the parallelogram law to $a = x - y_0$, $b = x - y_0'$: $\|y_0 - y_0'\|^2 = 2d^2 + 2d^2 - 4\|x - \frac{y_0+y_0'}{2}\|^2 \leq 4d^2 - 4d^2 = 0$, so $y_0 = y_0'$. □

Theorem 3.2 (Projection Theorem)**.** *Let H be a Hilbert space and $M \subset H$ a closed subspace. For every $x \in H$, there exist unique elements $m \in M$ and $z \in M^\perp$ such that*

$$x = m + z.$$

The element m is the unique best approximation of x in M, and is characterized by the orthogonality condition $x - m \in M^\perp$.

Proof. Since M is a closed convex set, Theorem 3.1 provides a unique $m \in M$ minimizing $\|x - m\|$. Set $z = x - m$.

Claim: $z \in M^\perp$. For any $v \in M$ and $\alpha \in \mathbb{F}$, $m + \alpha v \in M$ (since M is a subspace), so

$$\|z\|^2 \leq \|x - (m + \alpha v)\|^2 = \|z - \alpha v\|^2 = \|z\|^2 - 2\operatorname{Re}(\alpha\langle v, z\rangle) + |\alpha|^2\|v\|^2.$$

For $v \neq 0$, choose $\alpha = \langle z, v\rangle / \|v\|^2$. Then $\bar{\alpha} = \langle v, z\rangle / \|v\|^2$ and $\alpha\langle v, z\rangle = |\langle z, v\rangle|^2 / \|v\|^2$, giving

$$0 \leq -\frac{|\langle z, v\rangle|^2}{\|v\|^2},$$

hence $\langle z, v\rangle = 0$ for all $v \in M$, i.e., $z \in M^\perp$.

Uniqueness: If $x = m_1 + z_1 = m_2 + z_2$ with $m_i \in M$ and $z_i \in M^\perp$, then $m_1 - m_2 = z_2 - z_1 \in M \cap M^\perp$. But $\langle w, w\rangle = 0$ for any $w \in M \cap M^\perp$, so $w = 0$. □

Orthogonal Decomposition

If M is a closed subspace of a Hilbert space H, then $H = M \oplus M^\perp$ (internal direct sum), meaning every $x \in H$ has a unique decomposition $x = m + z$ with $m \in M$, $z \in M^\perp$. Moreover, $(M^\perp)^\perp = M$.

Definition 3.3 (Orthogonal Projection)**.** The *orthogonal projection* onto a closed subspace M of a Hilbert space H is the map $P_M \colon H \to H$ defined by $P_M(x) = m$, where $x = m + z$ is the unique decomposition from Theorem 3.2.

Proposition 3.4 (Properties of Orthogonal Projections)**.** *Let $P = P_M$ be the orthogonal projection onto a closed subspace M. Then:*

(i) *P is linear.*

(ii) *$P^2 = P$ (idempotent).*

(iii) $\operatorname{range}(P) = M$ *and* $\ker(P) = M^\perp$.

(iv) *$\langle Px, y\rangle = \langle x, Py\rangle$ for all $x, y \in H$ (self-adjoint).*

(v) *$\|P\| = 1$ if $M \neq \{0\}$ (bounded with operator norm 1).*

(vi) $\|x - Px\| \le \|x - m\|$ *for all* $m \in M$*, with equality if and only if* $m = Px$*.*

Proof. (i) If $x = m_1 + z_1$ and $y = m_2 + z_2$ are the decompositions, then $\alpha x + \beta y = (\alpha m_1 + \beta m_2) + (\alpha z_1 + \beta z_2)$, which is the decomposition since M and $M^\perp$ are subspaces. Thus $P(\alpha x + \beta y) = \alpha m_1 + \beta m_2 = \alpha Px + \beta Py$.

(ii) $Px \in M$, so $Px = Px + 0$ is the decomposition of Px, giving $P(Px) = Px$.

(iii) Immediate from the decomposition.

(iv) $\langle Px, y\rangle = \langle Px, Py + (I-P)y\rangle = \langle Px, Py\rangle + \langle Px, (I-P)y\rangle$. Since $Px \in M$ and $(I-P)y \in M^\perp$, the second term vanishes. Similarly $\langle x, Py\rangle = \langle Px, Py\rangle$.

(v) $\|Px\|^2 = \langle Px, Px\rangle = \langle Px, x\rangle \le \|Px\|\|x\|$ by Cauchy–Schwarz, so $\|Px\| \le \|x\|$, hence $\|P\| \le 1$. For $0 \ne m \in M$, $Pm = m$, so $\|P\| = 1$.

(vi) This is the best approximation property from Theorem 3.1. □

Projection Operators in TD Learning

In linear function approximation for RL, the value function is approximated in a subspace $M = \{\Phi w : w \in \mathbb{R}^d\}$ where Φ is a feature matrix. The LSTD algorithm computes an approximation $\hat{V} = P_M TV$ where T is the Bellman operator and P_M is the orthogonal projection onto M in a weighted L^2 norm. The properties above—boundedness ($\|P_M\| = 1$), idempotence, and self-adjointness—are essential for proving that the projected Bellman equation $V = P_M TV$ has a unique fixed point and that LSTD converges. The contraction factor of $P_M T$ is $\gamma\|P_M\| = \gamma < 1$.

4 Orthonormal Systems

Definition 4.1 (Orthonormal Set)**.** A set $\{e_\alpha\}_{\alpha\in A}$ in an inner product space is *orthonormal* if

$$\langle e_\alpha, e_\beta\rangle = \delta_{\alpha\beta} = \begin{cases} 1 & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \ne \beta. \end{cases}$$

Definition 4.2 (Fourier Coefficients)**.** Given an orthonormal set $\{e_\alpha\}_{\alpha\in A}$ and $x \in X$, the *Fourier coefficients* of x with respect to $\{e_\alpha\}$ are $\hat{x}_\alpha = \langle x, e_\alpha\rangle$.

Theorem 4.3 (Gram–Schmidt Process)**.** *Let* $(x_n)_{n=1}^\infty$ *be a linearly independent sequence in an inner product space* X*. Define*

$$e_1 = \frac{x_1}{\|x_1\|},$$
$$\tilde{e}_n = x_n - \sum_{k=1}^{n-1} \langle x_n, e_k\rangle\, e_k, \qquad e_n = \frac{\tilde{e}_n}{\|\tilde{e}_n\|} \quad (n \ge 2).$$

Then (e_n) *is an orthonormal sequence with* $\operatorname{span}\{e_1, \dots, e_n\} = \operatorname{span}\{x_1, \dots, x_n\}$ *for every* n*.*

Proof. By induction. The base case $n = 1$ is clear: $\|e_1\| = 1$.

Suppose $e_1, \dots, e_{n-1}$ are orthonormal with $\operatorname{span}\{e_1, \dots, e_{n-1}\} = \operatorname{span}\{x_1, \dots, x_{n-1}\}$. Then $\tilde{e}_n = x_n - \sum_{k=1}^{n-1}\langle x_n, e_k\rangle\, e_k$ satisfies, for $j = 1, \dots, n-1$:

$$\langle \tilde{e}_n, e_j\rangle = \langle x_n, e_j\rangle - \langle x_n, e_j\rangle = 0.$$

Since (x_k) is linearly independent, $x_n \notin \operatorname{span}\{x_1, \dots, x_{n-1}\}$, so $\tilde{e}_n \ne 0$. Setting $e_n = \tilde{e}_n/\|\tilde{e}_n\|$ gives $\|e_n\| = 1$ and $\langle e_n, e_j\rangle = 0$ for $j < n$.

The span equality follows since $x_n = \tilde{e}_n + \sum_{k=1}^{n-1}\langle x_n, e_k\rangle\, e_k$ expresses x_n in terms of $e_1, \dots, e_n$, and conversely e_n is a linear combination of $x_1, \dots, x_n$. □

Theorem 4.4 (Bessel's Inequality)**.** *Let $\{e_n\}_{n=1}^{\infty}$ be an orthonormal sequence in an inner product space X. For every $x \in X$,*

$$\sum_{n=1}^{\infty} |\langle x, e_n\rangle|^2 \leq \|x\|^2.$$

Proof. For any $N \in \mathbb{N}$, let $s_N = \sum_{n=1}^{N} \langle x, e_n\rangle\, e_n$. Then

$$\begin{aligned} 0 \leq \|x - s_N\|^2 &= \|x\|^2 - 2\,\mathrm{Re}\sum_{n=1}^{N} |\langle x, e_n\rangle|^2 + \sum_{n=1}^{N} |\langle x, e_n\rangle|^2 \\ &= \|x\|^2 - \sum_{n=1}^{N} |\langle x, e_n\rangle|^2, \end{aligned}$$

where we used orthonormality to compute $\|s_N\|^2 = \sum_{n=1}^{N} |\langle x, e_n\rangle|^2$ (Pythagorean theorem) and expanded the cross terms. Rearranging gives $\sum_{n=1}^{N} |\langle x, e_n\rangle|^2 \leq \|x\|^2$. Since this holds for all N, the series converges and the inequality persists. □

Bessel's Inequality – Energy Bound

The sum of squared Fourier coefficients is bounded by the squared norm of the signal. In the language of signal processing, the "energy" in the Fourier coefficients cannot exceed the energy of the original signal. Equality holds (Parseval's identity) if and only if the orthonormal set is complete.

5 Parseval's Identity and Fourier Expansion

Definition 5.1 (Orthonormal Basis (Hilbert Basis))**.** An orthonormal set $\{e_\alpha\}_{\alpha\in A}$ in a Hilbert space H is an *orthonormal basis* (or *complete orthonormal system*, or *Hilbert basis*) if $\overline{\mathrm{span}\{e_\alpha : \alpha \in A\}} = H$. Equivalently, $\{e_\alpha\}^{\perp} = \{0\}$.

Theorem 5.2 (Parseval's Identity and Fourier Expansion)**.** *Let $\{e_n\}_{n=1}^{\infty}$ be an orthonormal basis of a Hilbert space H. Then for every $x \in H$:*

(i) ***Fourier expansion:*** *$x = \sum_{n=1}^{\infty} \langle x, e_n\rangle\, e_n$ (convergence in norm).*

(ii) ***Parseval's identity:*** *$\|x\|^2 = \sum_{n=1}^{\infty} |\langle x, e_n\rangle|^2$.*

(iii) ***Generalized Parseval:*** *$\langle x, y\rangle = \sum_{n=1}^{\infty} \langle x, e_n\rangle\, \overline{\langle y, e_n\rangle}$ for all $x, y \in H$.*

Proof. Let $s_N = \sum_{n=1}^{N} \langle x, e_n\rangle\, e_n$.

(i) We show $s_N \to x$. Let $M_N = \mathrm{span}\{e_1, \ldots, e_N\}$. By Proposition 3.4, $s_N = P_{M_N} x$ (the projection onto M_N), since $x - s_N \perp e_n$ for $n = 1, \ldots, N$. For any $\varepsilon > 0$, since $\overline{\mathrm{span}\{e_n\}} = H$, there exist N_0 and scalars $c_1, \ldots, c_{N_0}$ with $\|x - \sum_{n=1}^{N_0} c_n e_n\| < \varepsilon$. For $N \geq N_0$, the best approximation property gives $\|x - s_N\| \leq \|x - \sum_{n=1}^{N_0} c_n e_n\| < \varepsilon$. Thus $\|x - s_N\| \to 0$.

(ii) By the Pythagorean theorem (since $s_N \perp (x - s_N)$):

$$\|x\|^2 = \|s_N\|^2 + \|x - s_N\|^2.$$

Since $\|x - s_N\| \to 0$ by part (i), we obtain $\|x\|^2 = \lim_{N\to\infty} \|s_N\|^2 = \sum_{n=1}^{\infty} |\langle x, e_n\rangle|^2$.

(iii) By the continuity of the inner product (Lesson 19, Proposition 4.1) and part (i):

$$\langle x, y\rangle = \Big\langle \sum_{n=1}^{\infty} \langle x, e_n\rangle\, e_n,\ \sum_{m=1}^{\infty} \langle y, e_m\rangle\, e_m \Big\rangle = \sum_{n=1}^{\infty} \langle x, e_n\rangle\, \overline{\langle y, e_n\rangle},$$

where the last equality uses orthonormality and the continuity of the inner product to interchange the inner product with the limit. □

Parseval's Identity – Isometry of Coefficient Map

The Fourier coefficient map $x \mapsto (\langle x, e_n \rangle)_{n=1}^{\infty}$ is an isometric isomorphism from H onto ℓ^2. That is, every separable Hilbert space is isometrically isomorphic to ℓ^2: up to isomorphism, there is essentially one infinite-dimensional separable Hilbert space.

Fourier Features in RL Function Approximation

In kernel-based RL, a positive definite kernel $k(x, y) = \int e^{i\omega^\top (x-y)} p(\omega)\, d\omega$ (Bochner's theorem) can be approximated by truncating the Fourier expansion: sample $\omega_1, \ldots, \omega_D \sim p$ and define random features $\phi(x) = (\cos(\omega_j^\top x), \sin(\omega_j^\top x))_{j=1}^{D} / \sqrt{D}$. Parseval's identity guarantees that the approximation error $|k(x, y) - \phi(x)^\top \phi(y)|$ vanishes as $D \to \infty$, since the tail of the Fourier series has vanishing energy.

6 Separability and Orthonormal Bases

Proposition 6.1 (Separability and Countable Orthonormal Bases)**.** *A Hilbert space is separable if and only if it admits a countable orthonormal basis.*

Proof. ($\Leftarrow$) If $\{e_n\}_{n=1}^{\infty}$ is a countable orthonormal basis, then the set of finite linear combinations of the e_n with rational (or rational complex) coefficients is countable and dense in H (by the Fourier expansion theorem: approximate any x by a truncation of its Fourier series, then approximate the Fourier coefficients by rationals).

($\Rightarrow$) Let $D = \{d_n\}$ be a countable dense subset. Extract a maximal linearly independent subsequence and apply the Gram–Schmidt process (Theorem 4.3). The resulting orthonormal sequence (e_n) satisfies $\overline{\operatorname{span}\{e_n\}} \supseteq \overline{\operatorname{span}\{d_n\}} \supseteq \overline{D} = H$, so (e_n) is an orthonormal basis. $\square$

Example 6.2 (Standard Orthonormal Basis of ℓ^2)**.** The standard unit vectors $e_n = (0, \ldots, 0, 1, 0, \ldots)$ (with 1 in position n) form an orthonormal basis of ℓ^2. The Fourier expansion of $x = (x_1, x_2, \ldots) \in \ell^2$ is simply $x = \sum_{n=1}^{\infty} x_n e_n$, and Parseval's identity reads $\|x\|^2 = \sum_{n=1}^{\infty} |x_n|^2$.

7 The Riesz Representation Theorem

Theorem 7.1 (Riesz Representation Theorem)**.** *Let H be a Hilbert space and $f\colon H \to \mathbb{F}$ a bounded linear functional ($f \in H^*$). Then there exists a unique $y \in H$ such that*

$$f(x) = \langle x, y \rangle \quad \text{for all } x \in H.$$

Moreover, $\|f\|_{H^*} = \|y\|_H$.

Proof. **Existence.** If $f = 0$, take $y = 0$. Suppose $f \neq 0$. Let $M = \ker f = \{x \in H : f(x) = 0\}$, which is a closed subspace (since f is continuous). Since $f \neq 0$, $M \neq H$.

By the Projection Theorem (Theorem 3.2), $H = M \oplus M^\perp$. We claim $\dim(M^\perp) = 1$: if $z_1, z_2 \in M^\perp \setminus \{0\}$, then $f(z_2)z_1 - f(z_1)z_2 \in M$ (check: $f(f(z_2)z_1 - f(z_1)z_2) = f(z_2)f(z_1) - f(z_1)f(z_2) = 0$) and also in $M^\perp$, so it equals 0. Thus z_1 and z_2 are linearly dependent, confirming $\dim(M^\perp) = 1$.

Hence $M^\perp = \operatorname{span}\{z\}$ for some $z \neq 0$ with $f(z) \neq 0$. Every $x \in H$ can be written $x = m + \alpha z$ with $m \in M$, so $f(x) = \alpha f(z)$.

Also, $\langle x, z \rangle = \langle m + \alpha z, z \rangle = \alpha \|z\|^2$ (since $m \perp z$), giving $\alpha = \langle x, z \rangle / \|z\|^2$.

Therefore $f(x) = \frac{f(z)}{\|z\|^2} \langle x, z \rangle = \langle x, \overline{\frac{f(z)}{\|z\|^2}}\, z \rangle$. Set $y = \overline{\frac{f(z)}{\|z\|^2}}\, z$.

Uniqueness. If $\langle x, y_1 \rangle = \langle x, y_2 \rangle$ for all x, then $\langle x, y_1 - y_2 \rangle = 0$ for all x. Taking $x = y_1 - y_2$ gives $\|y_1 - y_2\|^2 = 0$.

Norm equality. $|f(x)| = |\langle x, y\rangle| \leq \|x\|\|y\|$ by Cauchy–Schwarz, so $\|f\| \leq \|y\|$. Taking $x = y$: $f(y) = \langle y, y\rangle = \|y\|^2$, so $\|f\| \geq |f(y/\|y\|)| = \|y\|$. □

Riesz Representation – A Hilbert Space is Self-Dual

The map $y \mapsto \langle \cdot, y\rangle$ is a conjugate-linear isometric isomorphism $H \cong H^*$. A Hilbert space is thus "self-dual" (up to conjugation). This is a property unique to Hilbert spaces among Banach spaces: for L^p with $p \neq 2$, the dual $(L^p)^* = L^q$ (where $1/p+1/q = 1$) is a *different* space.

Riesz Representation and Kernel Methods

In a reproducing kernel Hilbert space (RKHS) $\mathcal{H}_K$, the evaluation functional $\delta_x \colon f \mapsto f(x)$ is bounded and linear. By the Riesz Representation Theorem, there exists $K_x \in \mathcal{H}_K$ with $f(x) = \langle f, K_x\rangle$. The function $K(x, y) = \langle K_x, K_y\rangle$ is the reproducing kernel. This is the foundation of kernel methods in ML: support vector machines, Gaussian processes, and kernel-based policy evaluation in RL all rest on this identification. The representer theorem—that the optimal solution to a regularized empirical risk problem lies in the span of the kernel evaluations—is a direct consequence.

8 Exercises

Exercise 8.1 (Projection Characterization (*)). Let $P \colon H \to H$ be a bounded linear operator on a Hilbert space. Show that P is an orthogonal projection onto some closed subspace if and only if $P^2 = P$ and $P^* = P$ (i.e., P is idempotent and self-adjoint). Identify the subspace.
Hint: Show $H = \mathrm{range}(P) \oplus \ker(P)$ and verify orthogonality using $P^* = P$.

Exercise 8.2 (Projection onto a Polynomial Subspace (*)). Let $H = L^2([0, 1])$ and $M = \mathrm{span}\{1, t\}$. Compute the orthogonal projection of $f(t) = t^2$ onto M by solving the normal equations $\langle f - P_M f, g\rangle = 0$ for all $g \in M$. Verify that the residual $f - P_M f$ is orthogonal to both 1 and t.

Exercise 8.3 (Gram–Schmidt and Legendre Polynomials (**)). Apply the Gram–Schmidt process to $\{1, t, t^2, t^3\}$ in $L^2([-1, 1])$ with inner product $\langle f, g\rangle = \int_{-1}^{1} f(t)\, g(t)\, dt$. Show that the resulting orthogonal polynomials are scalar multiples of the Legendre polynomials $P_0(t) = 1$, $P_1(t) = t$, $P_2(t) = \frac{1}{2}(3t^2 - 1)$, $P_3(t) = \frac{1}{2}(5t^3 - 3t)$.

Exercise 8.4 (Bessel's Inequality is Tight (**)). Give an example of an inner product space X, an orthonormal sequence (e_n) in X, and an element $x \in X$ such that Bessel's inequality is strict: $\sum_{n=1}^{\infty} |\langle x, e_n\rangle|^2 < \|x\|^2$. Explain why this cannot happen in a Hilbert space with an orthonormal *basis*.
Hint: Consider the space of finite sequences with the ℓ^2 inner product (which is not complete).

Exercise 8.5 (Riesz Representative of a Trace Functional (**)). Let $H = \mathbb{R}^{n\times n}$ with the Frobenius inner product $\langle A, B\rangle_F = \mathrm{tr}(B^\top A)$. The functional $f(A) = \mathrm{tr}(A)$ is linear and bounded. Find the Riesz representative $Y \in \mathbb{R}^{n\times n}$ such that $f(A) = \langle A, Y\rangle_F$ for all A, and verify $\|f\| = \|Y\|_F$.

Exercise 8.6 (The Dual of ℓ^2 via Riesz (**)). Let $f \in (\ell^2)^*$ be defined by $f(x) = \sum_{n=1}^{\infty} a_n x_n$ for a fixed sequence $(a_n) \in \ell^2$. Show that f is bounded with $\|f\| = \|(a_n)\|_{\ell^2}$, and identify the Riesz representative. Use this to give a direct proof that $(\ell^2)^* \cong \ell^2$.

Exercise 8.7 (Projection Error Bound in RL (***))**.** Let μ be a probability distribution on a finite state space $\mathcal{S}$, and let $\Phi \in \mathbb{R}^{|\mathcal{S}|\times d}$ be a feature matrix with rank d. Define the weighted inner product $\langle f, g\rangle_\mu = \sum_s \mu(s)f(s)g(s)$ and the projection Π_μ onto range(Φ) in this inner product. Let T be the Bellman evaluation operator with discount $\gamma \in (0, 1)$.

(a) Show that $\Pi_\mu T$ is a γ-contraction in $\|\cdot\|_\mu$ (use $\|\Pi_\mu\| = 1$ and the fact that T is a γ-contraction in $\|\cdot\|_\infty$; you may use the bound $\|Tv\|_\mu \le \|Tv\|_\infty$).

(b) Deduce that the projected Bellman equation $\hat{V} = \Pi_\mu T\hat{V}$ has a unique solution in range(Φ).

(c) Show the approximation error satisfies $\|\hat{V} - V^*\|_\mu \le \frac{1}{1-\gamma}\|V^* - \Pi_\mu V^*\|_\mu$, where V^* is the true value function.

9 Summary

- **Best approximation:** In a Hilbert space, every nonempty closed convex set has a unique nearest point to any given element.
- **Projection theorem:** For a closed subspace M, $H = M \oplus M^\perp$, and the projection P_M is the unique bounded, linear, idempotent, self-adjoint operator with range$(P_M) = M$ and $\|P_M\| = 1$.
- **Gram–Schmidt:** Any linearly independent sequence can be orthonormalized while preserving spans.
- **Bessel's inequality:** $\sum |\langle x, e_n\rangle|^2 \le \|x\|^2$ for any orthonormal sequence.
- **Parseval's identity:** For an orthonormal basis, $\|x\|^2 = \sum |\langle x, e_n\rangle|^2$ and $x = \sum \langle x, e_n\rangle\, e_n$.
- **Separability:** A Hilbert space is separable if and only if it has a countable orthonormal basis, and all separable infinite-dimensional Hilbert spaces are isomorphic to ℓ^2.
- **Riesz Representation:** Every $f \in H^*$ is of the form $f(x) = \langle x, y\rangle$ for a unique y with $\|f\| = \|y\|$. This gives the isometric isomorphism $H \cong H^*$.
- **RL connections:** The projection theorem underlies LSTD and projected Bellman equations; Parseval's identity justifies Fourier feature approximations; the Riesz theorem is the foundation of RKHS theory and gradient identification.

10 Further Reading

For a comprehensive treatment of Hilbert space theory, see Conway [1990], Chapters I–II, and Kreyszig [1989], Chapter 3. The connection between projections and approximation theory is developed in detail in Brézis [2011], Chapter 5. For applications of RKHS theory to machine learning, see the survey by Berlinet and Thomas-Agnan (2004).

The next lesson (Lesson 21) develops the fundamental theorems of functional analysis—Hahn–Banach, Uniform Boundedness, Open Mapping, and Closed Graph—which hold in general Banach spaces rather than requiring the inner product structure.

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